



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure and Syllabus

(From Academic Session 2024-25 onwards)

B. TECH
MECHANICAL ENGINEERING
4th SEMESTER



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Course Structure

(From Academic Session 2024-25 onwards)

B. Tech 4th Semester: Mechanical Engineering

Semester IV

Sl. No.	Course Code	Course Type	Course Title	Hours per Week			Credit	Marks	
				L	T	P	C	CE	ESE
Theory									
1	ME241401	PCC	Applied Thermodynamics	3	0	0	3	30	70
2	ME241402	PCC	Dynamics of Machines	3	0	0	3	30	70
3	ME241403	PCC	Machine Drawing	2	0	2	3	30	70
4	ME241404	PCC	Material Science and Engineering	3	0	0	3	30	70
5	ME241405	PCC	Fluid Mechanics-I	3	0	0	3	30	70
6	HS241406	HSMC	Finance and Accounting	3	0	0	3	30	70
7	AU241407	AU	Environmental Science	3	0	0	0 (PP/NP)	100	-
Practical									
8	ME241411	PCC	Applied Thermodynamics Lab	0	0	2	1	15	35
9	ME241412	PCC	Dynamics of Machine Lab	0	0	2	1	15	35
10	ME241414	PCC	Material Science and Engineering Lab	0	0	2	1	15	35
11	PR241415	PR	Micro Project (Skill Based)	0	0	4	2	15	35
TOTAL				20	0	12	23	340	560
Total Contact Hours per week: 32									
Total Credit: 23									

Course Code	Course Title	Hours per week L-T-P	Credit
ME241401	Applied Thermodynamics	3-0-0	3

Prerequisites: Basic Thermodynamics

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Determine the amount of air required for the combustion of solid, liquid and gaseous of fuels for the given composition of the fuel, the percentage of excess air supply and the amount of combustion gases produced.
- **CO2:** Explain the working principles of SI and CI engine, 2-stroke and 4-stroke engines, and air standard cycles for SI and CI engines.
- **CO3:** Estimate output, axial thrust, diagram efficiency, stage efficiency, overall efficiency, turbine blade height of steam turbine and efficiency of condensers.
- **CO4:** Analyze compressed air systems by evaluating various performance parameters, while gaining a comprehensive understanding of key concepts, including the types, performance characteristics, and applications of different compressors (reciprocating, rotary, axial flow, and centrifugal).
- **CO5:** Analyze gas turbine systems by estimating key performance metrics such as thermal efficiency, specific power consumption, and the power output generated by the gas turbine unit, based on the provided operating conditions.
- **CO6:** Analyze a vapor compression refrigeration system by estimating key performance parameters such as the refrigerating effect, refrigeration capacity, coefficient of performance (COP), and the power required to operate the system under specified operating conditions.

MODULE 1: Gas Power Cycle (6 Lectures)

An overview of reciprocating IC engines, working principle of 2-stroke and 4-stroke engines, Air-standard cycles for IC engines. An overview of gas turbine, its classification, air-standard cycle for gas turbine.

MODULE 2: Combustion Processes (6 Lectures)

Fuels and combustion, theoretical and actual combustion processes, combustion equation, enthalpy of formation and enthalpy of combustion, first law analysis of reacting systems, adiabatic flame temperature.

MODULE 3: Steam Turbine (6 Lectures)

Impulse and Reaction turbines, velocity diagrams, output, axial thrust, diagram efficiency, stage efficiency, overall efficiency, turbine blade height, Illustrative problems reheat factor, condition curve, compounding of Impulse turbine, Condenser, classification and vacuum efficiency.

MODULE 4: Air Compressors (8 Lectures)

Application of compressed air, Compressed air system, types of compressors, reciprocating compressor, capacity of compressor, compressor performance (isothermal efficiency, volumetric efficiency), effect of clearance volume, multistage compression with intercooling, ideal intermediate pressure, elementary constructional details, rotary compressors, applications, axial flow compressor, centrifugal compressor, characteristics, charging and chocking.

MODULE 5: Gas Turbine Cycle (8 Lectures)

Open and closed cycle, Air standard gas turbine cycle (Brayton cycle), methods of improving thermal efficiencies of gas turbine, optimal performance of various cycles, combined gas and steam cycles, cogeneration combustion chambers;

Basic Concept of Jet Propulsion: turbojet, turboprop, turbofan, ramjet, thrust and propulsive efficiency; Rocket Propulsion

MODULE 6: Refrigeration and Psychrometric Process (8 Lectures)

Definition of COP, refrigerating effect, COP analysis of Simple vapour compression refrigeration cycle, Vapour absorption system ($\text{LiBr} + \text{H}_2\text{O}$; $\text{NH}_3 + \text{H}_2\text{O}$), refrigerants.

Psychrometry properties, Psychrometric charts and basic Psychrometric processes.

Text Books/References:

1. P.K. Nag, Engineering Thermodynamics, McGraw Hill Publications, 6th edition 2017
2. Cengel and Boles, Thermodynamics – An Engineering Approach, Tata McGraw Hill Publications., 8th edition 2017
3. R. K. Rajput, Thermal Engineering, Laxmi Publications Ltd., 6th edition 2016
4. S. Domkundwar, C.P. Kothandaraman, Anand Domkundwar, A Course in Thermal Engineering, Dhanpat Rai and Co., 2017
5. C.P. Arora, Refrigeration and Air Conditioning, Tata McGraw Hill, 3rd edition 2017
6. R.S.Khurmi, A Textbook of Refrigeration & Air condition, S. Chand, 2019
7. W.F. Stoecker and J.F. Jones, Refrigeration and Air Conditioning, McGraw Hill International Edition, 2nd edition

Course Code	Course Title	Hours per week L-T-P	Credit
ME241402	Dynamics of Machines	3-0-0	3

Prerequisites: Theory of Machines

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Analyze velocity and acceleration in mechanisms by utilizing kinematic principles.
- **CO2:** Assess the performance of governors in controlling speed fluctuations and maintaining stability in mechanical systems.
- **CO3:** Evaluate stability and control in rotating systems by using gyroscopic principles.
- **CO4:** Apply balancing techniques for rotating masses to balance several masses rotating in same plane and different planes.
- **CO5:** Apply balancing techniques for reciprocating masses to minimize vibrations, improve stability, and enhance the efficiency of mechanical systems and engines.

MODULE 1: Analysis & Synthesis of Mechanism (8 Lectures)

Relative velocity method for finding velocity and acceleration in link, slider crank mechanism, and four bar chain; Coriolis acceleration component, rubbing velocity at pin joint, forces acting in a mechanism, mechanical advantage, Davis and Ackermann steering gear, universal Hook joint, synthesis of mechanism, analysis vs. synthesis of mechanism, number and dimensional synthesis, function generation by graphical method for two position and three position synthesis, Freudenstein's equation, Chebyshev spacing method for precision points.

MODULE 2: Governors & their performance (8 Lectures)

Function and types of governors, Watt, Porter, and Proell governor; Hartnell, Wilson-Hartnell, and Hartung spring-controlled governors, sensitiveness, stability, isochronism, hunting, effort & power, controlling force, stability of spring-controlled governor, friction and insensitiveness.

MODULE 3: Gyroscopic Effects & Applications (8 Lectures)

Concept of spinning and precession, gyroscopic couple, effect of gyroscopic couple on rotor on a shaft, airplane, naval ship, four-wheeler automotive vehicle, and two-wheeler vehicle; gyroscope.

MODULE 4: Balancing of Rotating Masses (8 Lectures)

Need of balancing, balancing of single rotating mass in same plane and in different plane, analytical and graphical methods for balancing of several masses rotating in same plane and rotating in different planes.

MODULE 5: Balancing of Reciprocating Masses (10 Lectures)

Balancing of reciprocating engine, partial balancing of primary force, partial balancing of locomotives, variation of tractive force, swaying couple, hammer blow, primary balance of multi cylinder in-line engine, secondary balance of multi cylinder in-line engine, V-engine balancing, method of direct and reverse cranks.

Text Books/References:

1. Dr. R.K. Bansal, Dr. J.S. Brar, Theory of Machines, Laxmi Publications(P) Ltd., 5th edition, 2016
2. Ratan S.S., Theory of Machines, McGraw Hill Education, 4th edition, 2017
3. Prof. P. L. Ballaney, Theory of Machines and Mechanisms, Khanna Publishers, 25th edition, 2014
4. Sadhu Singh, Theory of Machines, Pearson Education India, 3rd edition 2011
5. John J. Uicker, Jr., Gordon R. Pennock and Joseph E. Shigley, Theory of Machines and Mechanisms, Oxford University Press, 5th edition 2017

Course Code	Course Title	Hours per week L-T-P	Credit
ME241403	Machine Drawing	2-0-2	3

Prerequisites: NIL

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Illustrate the Orthographic and Sectional views of machine components.
- **CO2:** Draw the profiles of thread and locking devices such as nuts and bolts.
- **CO3:** Sketch various joints-key and cotter joints, riveted joints, welded joints, coupling, pipe joints as well as machine elements.
- **CO4:** Assemble the components of engine and valve parts such as piston, stuffing box, cross head, connecting rod, eccentric, feed check valve Illustrate, stop valve, blow off cock, non-return valve.
- **CO5:** Understand the symbols and notations used in the production drawing.

MODULE 1: Orthographic and Sectional views (6 Lectures)

Introduction, 1st and 3rd angle projection, Front, Top and Side views of a machine component, Sectional views, Half-sectional views of machine components.

MODULE 2: Screw Fasteners (6 Lectures)

Introduction, Screw Thread Nomenclature, Forms of Thread, Thread Profiles- V-Thread, Buttress Thread, British Standard Whitworth (B.S.W) Thread, Square Thread ACME Thread, Worm Thread, Thread designation, Multi-start Thread, left hand thread, right hand thread, Locking devices for nuts, Different types of bolts and nuts.

MODULE 3: Keys, Joints and couplings (8 Lectures)

Introduction, Saddle Key, Sunk Key, Cotter Joint with Sleeve, Socket and Spigot Ends, and with Gibs, Knuckle joints. Rigid Coupling Flanged Coupling, Sleeve or Muff Coupling, Flexible Coupling, Pipe Joints, Hydraulic Joints.

MODULE 4: Riveted and Welded Joints (6 Lectures)

Introduction, Rivet heads, Definitions-Pitch, Margin, Chain Riveting, Zig-Zag Riveting, welded joints and Symbols, Dimensioning of welds.

MODULE 5: Assembly Drawings (10 Lectures)

- Engine Parts: Stuffing Box, Crosshead, Connecting Rod, Eccentric, Piston
- Valves: Stop Valve, Feed Check Valve, blow off Cock, Non-Return Valve

MODULE 6: Production Drawing (6 Lectures)

Introduction, Method and placing limit dimensions, Datum symbols, Tolerance of forms and positions, Surface roughness machining symbols, Direction of lay symbols on production or working drawing

Text Books/References:

1. N.D Bhatt, Machine Drawing, Charotar Publishing House Pvt. Ltd., 51st edition 2022
2. K. L. Narayana, P. Kanniah, K. Venketa Reddy, Machine Drawing, New Age International Pvt. Ltd, 5th edition, 2016
3. N. Sidheswar, P. Kanniah and V.V.S. Sastry, Machine Drawing, McGraw Hill Education, 3rd edition 2017
4. K. L. Narayana, Production Drawing, New Age International Pvt. Ltd, 3rd edition 2014

Course Code	Course Title	Hours per week L-T-P	Credit
ME241404	Material Science and Engineering	3-0-0	3

Prerequisites: NIL

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Analyse the failure of metals by studying the mechanical properties and different failure mechanism like fracture, fatigue and creep.
- **CO2:** Identify crystal structures and imperfections for various materials.
- **CO3:** Analyse binary phase diagrams of alloys including Fe-Fe₃C system
- **CO4:** Design proper heat treatment process to tailor material properties for ferrous alloys.
- **CO5:** Classify different Engineering alloys, namely Copper, Aluminium, Cast iron, stainless steel and non-metallic materials.

MODULE 1: Mechanical Property measurement (8 Lectures)

Tensile, compression and torsion tests, relations between true and engineering stress-strain curves, yielding and yield strength, ductility, resilience, toughness, Hardness: Rockwell, Brinell and Vickers and their relation to strength

Fracture of Metals: Ductile fracture, Brittle fracture, Ductile and Impact Testing, Toughness and Impact testing

Fatigue of Metals: Cyclic stress, SN curve, endurance and fatigue limits, Fatigue life calculations

Creep and Stress rupture of metals: Creep of metals, Creep test

MODULE 2: Crystal Structures and Imperfection in solids (8 Lectures)

Crystal Structures: Crystal systems and Bravais Lattice, Principal Metallic Crystal Structure (SC, BCC, FCC, HCP), Miller Indices and direction for Crystallographic Planes in Cubic unit cell, Crystal structure analysis by X ray diffraction

Imperfection in solids: Point, line, interfacial and volume defects; deformation by twinning, Stacking faults.

MODULE 3: Phases and Phase Diagram (8 Lectures)

Phase diagram of pure substance, Gibbs Phase rule, Cooling curves, Lever rule, Binary phase diagram: Eutectic, Eutectoid, Peritectic, Peritectoid; Iron-Carbon equilibrium diagram, Microstructure changes during phase transformation.

MODULE 4: Heat Treatment of Plain Carbon Steel (6 Lectures)

Transformation in plain carbon steel: TTT diagram, Effect of Alloying elements in Alloy Steel, Different heat treatment processes: Hardening, Tempering, Annealing, Normalizing, Case Hardening, Cyaniding, Nitriding, Flame and Induction Hardening.

MODULE 5: Engineering Alloys (6 Lectures)

Cast iron: General Properties, Types of cast iron, Stainless steel: Ferritic, Martensitic and Austenitic stainless steel, Copper Alloy: General properties of copper, Production of Copper, Classification of Copper Alloy, Wrought Copper alloy, Aluminium Alloy: General properties of Aluminium and its production, Wrought Aluminium Alloys, Aluminium Casting Alloys

MODULE 6: Non-Metallic Materials (6 Lectures)

Classification of non-metallic materials, Rubber: classification, Properties, processing and applications, Plastics: Thermosetting and Thermoplastics, General purpose Thermoplastics (Polyethylene, Polypropylene, Polystyrene), Engineering thermoplastics (Nylons, Polycarbonate, Acetals), Ceramics: Properties and applications, Processing of Ceramics, Glasses: Glass transition temperature, Structure and composition of glasses, Composites: Properties and applications, Fibers for Reinforced Plastic Composite Materials (Glass fibers, carbon fibers)

Text Books/References:

1. William D. Callister Jr. and David G. Rethwisch, Materials Science and Engineering: An Introduction, Wiley, 9th Edition 2013
2. James F. Shackelford, Introduction to Materials Science for Engineers, Pearson, 8th Edition 2020
3. William F. Smith and Javad Hashemi, Foundations of Materials Science and Engineering, McGraw-Hill Education, 8th Edition 2020
4. S.L Kakani and Amit Kakani, Material Science, New Age International Private Limited, 2016
5. V. Raghavan, Materials Science and Engineering, Prentice Hall India Learning Private Limited, 6th Edition 2015

Course Code	Course Title	Hours per week L-T-P	Credit
ME241405	Fluid Mechanics-I	3-0-0	3

Prerequisites: NIL

Course Outcome (CO): After completion of the course, the students will be able to:

- **CO1:** Apply conservation laws to fluid flow problems in engineering applications.
- **CO2:** Design experimental procedure for physical model studies.
- **CO3:** Compute drag and lift coefficients using the theory of boundary layer flows.
- **CO4:** Analyze and design free surface and pipe flows
- **CO5:** Formulate and solve one-dimensional compressible fluid flow problems

MODULE 1: Introduction (5 Lectures)

Flow properties-density, specific weight, specific volume, relative density, viscosity, Newton's law of viscosity, classification of fluids, continuum.

MODULE 2: Fluid Static's (8 Lectures)

Introduction, pressure at a point, Pascals law, Pressure variation in a fluid at rest, measurement and scales of pressure, pressure measuring devices, hydrostatic force on a horizontal, vertical, inclined planes and rough surface, buoyancy stability of floating and submerged bodies, metacentre and metacentric height, conditions of equilibrium of a floating and submerged bodies.

MODULE 3: Kinematics of Fluids (8 Lectures)

Flow field and description of fluid motion, velocity and acceleration, stream line pathline, streak line, stream tube, steady, unsteady flow uniform and non-uniform flow, rotational and irrotational flows, translation, vorticity, stream function, velocity potential function, flow net, source, sink, vortex flow, free and forced vortex, doublet, continuity and its analysis based on integral form.

MODULE 4: Dynamics of Fluid Flow (8 Lectures)

Euler's equation of motion, its vectorial approach, Euler's eq. along a streamline, analytical and vectorial approach of Bernoulli's equation, energy equation and momentum equation and its vectorial approach, application of Bernoulli's eq. to real fluid flow, dynamic forces on plain and curved surfaces due to impingement of liquid jets.

MODULE 5: Flow measurement (8 Lectures)

Concept of static and Stagnation pressure, application of pitot tube, hot wire anemometer, venturimeter, orificemeter, the phenomenon of jet contraction, hydraulic co-efficient of an orifice, factors affecting orifice coefficients.

MODULE 6: Dimensional analysis and Dynamic Similitude (5 Lectures)

Buckingham Pi theorem, dimensionless numbers, distorted models.

Text Books/References:

1. Ethirajan Rathakrishnan, Fluid Mechanics: An Introduction, PHI, June 2022
2. Frank M White, Fluid Mechanics, Mc Graw Hill, 7th edition
3. Dr RK Bansal, A Textbook of fluid Mechanics And hydraulic Machines, Laxmi Publication, 9th Edition
4. R.K.Rajput, A Textbook of Fluid Mechanics and Hydraulic Machines (In SI Units), S.Chand, 6th edition 2016

Course Code	Course Title	Hours per week L-T-P	Credit
HS241406	Finance and Accounting	3-0-0	3

Prerequisites: High School Education

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Develop a strong foundation in accounting principles, including the classification of accounts, the Double Entry System, the application of the Golden Rules of Debit and Credit, record financial transactions in journals, post them into ledgers, and prepare a Trial Balance.
- **CO2:** Understand the role of subsidiary books in accounting, including different types of cash books and their preparation; learn to reconcile differences between the Cash Book and Pass Book balances and prepare a Bank Reconciliation Statement
- **CO3:** Differentiate between capital and revenue expenditure, understand the concepts of bad debts and provisions for doubtful debts and gain the skills to prepare final accounts by incorporating necessary adjustments.
- **CO4:** Explore the concepts of savings and investment, including their benefits and differences, identify various investment avenues in physical and financial assets and understand the distinction between trading and investment to make informed financial decisions.
- **CO5:** Have a comprehensive understanding of financial markets, covering money and capital markets, key financial instruments, IPOs, stock exchanges, and mutual funds, and analyze the structure and functions of financial markets and assess different financial instruments used for investment.
- **CO6:** Acquire knowledge and skills in stock market investment and analysis, applying fundamental and technical analysis to evaluate stocks, learn to interpret stock market charts and patterns, assess stock price movements, and utilize various analytical tools for informed investment decisions.

MODULE 1: Introduction to Accounting

Concept and classification of Accounts, Transaction, Double Entry System of Book Keeping, Golden rules of Debit and Credit, Journal- Definition, Advantages, Procedure of Journalising, Ledger, Advantages, Rules regarding Posting, Balancing of ledger accounts, Trial Balance- Definition, Objectives, Procedure of preparation.

MODULE 2: Subsidiary Books, Cash Book and Bank Reconciliation

Name of Subsidiary Books, Cash Book- Definition, Advantages, Objectives, Types of Cash Book, Preparation of different types of Cash Books, Bank Reconciliation Statement, Reasons of disagreement between Cash Book with Pass Book balance, Preparation of bank Reconciliation Statement.

MODULE 3: Expenditures, Bad Debts and Final Accounts

Concept of Capital Expenditure and Revenue Expenditure, Bad Debts, Provision for Bad and Doubtful debts, Final Accounts: Preparation of Trading Account, Profit and Loss account and Balance Sheet with adjustments.

MODULE 4: Introduction to Saving and Investment

Savings: Concept, Benefits and saving alternatives, Disadvantage of Saving – Time Value of Money. Investment: Meaning, Advantage, Investment Avenues – Investment in Physical and financial assets, Difference between Saving and Investment; Trading- Meaning, Difference between Trading and Investment.

MODULE 5: Financial Market and Instruments.

Financial Markets and Instruments: Money Market, Capital Market, Money Market Instruments and features. Capital Markets and Features, New Issue Market and Stock Market; New Issue market- Initial Public Offering , Market Players, Secondary Market :Equity and Debentures, BSE and NSE, Index and its uses; Mutual Funds: Meaning and Types.

MODULE 6: Stock Market Investment and Analysis

Stock Market Investment: Fundamental Analysis- Economic Analysis-Industry Analysis and Technical Analysis - Tools of technical analysis, Understanding various types of charts and patterns- (Line chart, Bar Chart, Candlestick Chart Point and Figure Chart)

Textbooks/Reference Books

1. Dam, B. B.Sarda, R.A. Barman, R. Kalita, B., Theory and Practice of Accountancy- Vol. I, Gayatri Publication, 2019
2. Goel, D.K Goel, R. Goel,S. Accountancy, Avichal publishing company, 2020
3. Kevin, S., Security Analysis and Portfolio Management, Prentice Hall India., 2022
4. Pathak,B., Indian Financial System, Pearson Education, 2024
5. Chandra, P., The Invesment Game: How To win, Tata McGrawHill, 2012

Course Code	Course Title	Hours per week L-T-P	Credit
AU241407	Environmental Science	3-0-0	0

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Relate environment and its associated problems
- **CO2:** Explain basic ethics of living and protection of other organisms
- **CO3:** Applying sustainable development principles

MODULE 1: Introduction

Environment and Ecology, Objectives of ecological study, Aspects of Ecology, Autecology, Synecology; Ecosystem, Structural and functional attributes of an ecosystem, Food chain and foodweb, Energy flow, Bio geochemical cycles

MODULE 2: Land: Use and Abuse

Land use: Impact of land–use on environmental quality, Land degradation, Control of land degradation, Wasteland, Wet lands

MODULE 3: Water Pollution

Introduction: Water quality standards, Water pollution

MODULE 4: Air Pollution

Introduction, Air pollution system, Air pollutants, Air pollution laws

MODULE 5: Noise Pollution

Sources of noise pollution, Effects of noise, Physical effects, Physiological effects

MODULE 6: Control of Pollutions

Control of water pollution and management, Water pollution legislations, Water quality management in surface water; Control of air pollution, source correction method, Pollution control equipment; controls of Noise pollution

Textbooks/Reference Books

1. Dr Suresh Dhameja, Environmental engineering and management, 2010
2. Dr B.S. Chauhan, Environmental studies
3. Henry and Hence, Environmental science and engineering
4. Dr Susmitha Baskar, Environmental studies for undergraduate course
5. Clair Sawyer, Chemistry for environmental engineering and science

Course Code	Course Title	Hours per week L-T-P	Credit
ME241411	Applied Thermodynamics Lab	0-0-2	1

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Determine the performance of refrigeration and air conditioning systems
- **CO2:** Estimate the performance of reciprocating air compressor and IC engines
- **CO3:** Apply different equipment for measuring temperature, pressure, relative humidity, rotational speed and flow rate
- **CO4:** Examine the accuracy of thermocouple
- **CO5:** Estimate psychrometric properties of air, dryness fraction of steam, and the percentage of combustion gases and air supply

MODULE 1:

Determination of the coefficient of performance of a Vapour compression refrigeration system and Vapour absorption refrigeration. **(4 Lectures)**

MODULE 2:

Determination of the COP of an air-conditioning test rig. **(2 Lectures)**

MODULE 3:

Determination of the performance of IC engine. **(4 Lectures)**

MODULE 4:

Determination of volumetric efficiency, isothermal efficiency and compression ratio of reciprocating compressor. **(2 Lectures)**

MODULE 5:

Calibration of thermocouple. **(2 Lectures)**

MODULE 6:

Determination of dryness fraction of steam. **(2 Lectures)**

MODULE 7:

Analysis of combustion process. **(4 Lectures)**

Text Books/References:

1. P.K. Nag, Engineering Thermodynamics, McGraw Hill Publications, 6th edition 2017
2. R. K. Rajput, Thermal Engineering, Laxmi Publications Ltd., 6th edition 2016
3. C.P. Arora, Refrigeration and Air Conditioning, Tata McGraw Hill, 3rd edition 2017
4. R.S.Khurmi, A Textbook of Refrigeration & Air condition, S. Chand, 2019

Course Code	Course Title	Hours per week L-T-P	Credit
ME241412	Dynamics of Machine Lab	0-0-2	1

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Compute Coriolis component of acceleration
- **CO2:** Determine characteristics curves of for Watt, Porter, Proell and Hartnell governor for sleeve position against speed and radius of rotation against controlling force.
- **CO3:** Determine percentage variation of theoretical and experimental gyroscopic couple
- **CO4:** Determine Static Balance of a set of four blocks with unbalance by properly positioning them in angular and lateral position on the shaft.
- **CO5:** Plot the $x - \theta$ (follower displacement vs. angle of cam rotation) curves for different cam follower pairs.

MODULE 1:

Coriolis Component of Acceleration Apparatus. **(4 Lectures)**

MODULE 2:

Characteristics curves of for Watt, Porter, Proell and Hartnell governor for sleeve position against speed and radius of rotation against controlling force. **(4 Lectures)**

MODULE 3:

Percentage variation of theoretical and experimental gyroscopic couple. **(4 Lectures)**

MODULE 4:

Static Balance of a set of four blocks with unbalance by properly positioning them in angular and lateral position on the shaft. **(4 Lectures)**

MODULE 5:

Study the various cam and follower pairs, effect of follower weight on bounce the effect of spring compression on bounce. **(4 Lectures)**

Text Books/References:

1. Dr. R.K. Bansal, Dr. J.S. Brar, Theory of Machines, Laxmi Publications(P) Ltd., 5th edition, 2016
2. Ratan S.S., Theory of Machines, McGraw Hill Education, 4th edition, 2017
3. Prof. P. L. Ballaney, Theory of Machines and Mechanisms, Khanna Publishers, 25th edition, 2014

Course Code	Course Title	Hours per week L-T-P	Credit
ME241414	Material Science and Engineering Lab	0-0-2	1

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Compute the hardness of a flat mild steel and high carbon steel specimen
- **CO2:** Compute the impact resistance of a notched bar flexure specimen
- **CO3:** Compute the compressive strength of materials
- **CO4:** Conduct typical engineering microscopic observation to recognize the microstructure of cast iron metal.

MODULE 1: Hardness Test

To determine hardness of a flat mild steel and high carbon steel specimen by using Rockwell and Brinell Hardness Tester Machine. **(4 Lectures)**

MODULE 2: Charpy Pendulum Impact Test

To determine impact resistance of an assigned specimen in the form of a notched bar flexure specimen. **(4 Lectures)**

MODULE 3: Compressive Strength Test

To determine the compressive Strength of a given specimen. **(4 Lectures)**

MODULE 4:

Preparation of metallic specimen for metallographic study by grinding, polishing and etching. **(2 Lectures)**

MODULE 5:

Study of the Microstructures of Cast Iron. **(2 Lectures)**

Text Books/References:

1. William F. Smith and Javad Hashemi, Foundations of Materials Science and Engineering, McGraw-Hill Education, 2009
2. S.L Kakani and Amit Kakani, Material Science, New Age International Private Limited, 2016

Course Code	Course Title	Hours per week L-T-P	Credit
PR241415	Micro Project (Skill Based)	0-0-4	2

Prerequisites: Knowledge of basic science

Objectives:

- To identify problems based on societal or research needs
- To acquaint with the process of solving the problem in a group
- To apply engineering knowledge to attempt solutions to the problems

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Analyze engineering problems and their solutions
- **CO2:** Apply modern engineering tools to attempt solutions to the problems
- **CO3:** Adapt with computational thinking and automation skills
- **CO4:** Realize the need of professional communications, team work and self-learning

General Guidelines:

1. Students shall form a group of 3 to 4 students.
2. Each group shall conduct field survey or literature survey in consultation with faculty supervisor/mentor or internal project committee of faculty members to identify the needs of the society.
3. Each group shall formulate the problems on the basis of their survey and submit the implementation plan to the concerned faculty supervisor/mentor or the committee.
4. A log book is to be maintained by each group, wherein the group can record the work progress. The supervisor shall monitor the progress.
5. The faculty supervisor/mentor or the committee may give inputs to the students during the project activities, however, focus shall be on self-learning.
6. Students in a group shall understand problem effectively, propose multiple solutions and select the best solution in consultation with faculty supervisor/mentor or the committee.
7. Students shall convert the best solution into a working model/prototype using various components of their domain areas.
8. Students shall demonstrate and validate the model with proper justification and a report is to be compiled in a standard format/template

Assessment Criteria:

The Micro Project shall be assessed based on the following criteria:

Sl No.	Component	Weightage
1	Proper identification of the problem	20
2	Quality of survey	15
3	Clarity of problem definition	10
4	Presence of innovation	5
5	Effective use of modern engineering tools	10
6	Cost effectiveness	5
7	Societal impact	10
8	Functioning of the working model	15
9	Fulfilment of standard engineering norms	5
10	Clarity in written and oral communication	5
		100

Examinations:

Sl No.	Type of examination/evaluation	Marks (50)
1	Mid-term presentation on Progress	10
2	Demonstration of the project	10
3	End-term presentation	25
4	Viva-voce examination	5
